
UNIT 4 OPEN ECONOMY

MACROECONOMICS -I*

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4.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the national income identities in an open economy;
- assess the relation between aggregate expenditure and income in an open economy;
- explain the relation between private savings-investment gap, fiscal deficit and current account deficit;
- explain the concept of Balance of Payments (BOP);
- explain current and Capital accounts of BOP;
- explain the concepts of nominal and real exchange rates; and
- explain fixed and flexible exchange rate regimes.

4.1 INTRODUCTION

In this Unit we will examine the national income identities, with a view to identifying certain fundamental differences between open and closed economies. Thereafter we examine certain basic concepts relating to Balance of Payments (BOP) and exchange rates. The BOP accounts record the transactions of an economy with the 'rest of the world' and provide important insights into the pattern of international trade and capital flows. The exchange rate is also an important variable in an open economy with important implications for international trade and competitiveness.

4.1 NATIONAL INCOME IDENTITY IN AN OPEN ECONOMY

In an open economy domestic agents have manifold economic interactions with foreigners; e.g., goods are sold to, as well as purchased from foreigners. The sale of goods and services to foreigners are exports (X) of the country and purchases from foreigners constitute imports (M). In an open economy, aggregate expenditure on final goods by domestic agents (households, firms and

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government) given by $[C + I + G]$, includes expenditure on domestic as well as imported goods. Aggregate spending on *domestic* goods by domestic agents is given by $[C + I + G - M]$. To this we add expenditure on domestic goods by foreigners or exports, to obtain *total* final expenditure on domestically produced goods, i.e., $[C + I + G + X - M]$.

From the equivalence between the expenditure, income and product approaches to measurement of aggregate output, we can say that aggregate final expenditure is equal to aggregate output produced in the economy or GDP (Y).

So the national income identity for an open economy can be written as:

$$Y = C + I + G + X - M$$

$$\text{or, } Y = C + I + G + NX \text{ (where } NX = X - M \text{)} \quad (4.1)$$

Note that $(X - M)$ is also referred to as net exports (NX) or balance of trade. When exports exceed imports, NX is positive and there is a trade surplus; when imports exceed exports, NX is negative and there is a trade deficit.

In an open economy we draw a distinction between GDP and GNP (Gross National Product). While GDP refers to aggregate incomes earned (by domestic residents and foreigners) within the geographical boundaries of the country, GNP is defined as the aggregate income of citizens of the country, irrespective of whether they are located within the country or abroad. E.g., the salary of a German consultant located in New Delhi would be a part of German GNP and India's GDP; the salary of an Indian worker in Singapore would be a part of Singapore's GDP and India's GNP.

So we define net factor income from abroad (NFIA) as follows:

NFIA = Income earned abroad by domestic factors of production *minus* income earned by foreign factors of production in the domestic economy.

To obtain GNP, NFIA is added to GDP (denoted often as Y in macroeconomics):

$$\text{i.e., } GNP = Y + NFIA \quad (4.2)$$

Note that NFIA can be positive or negative and the difference between GDP and GNP can often be quite large. For example, GDP may exceed GNP ($NFIA < 0$) in countries where most production units are owned by foreign multinationals (so that a large share of the incomes generated within the economy, accrue to foreigners) or which have a sizeable migrant population employed abroad.

Using the national income identity we can show in an open economy aggregate expenditure ($C+I+G$) can exceed aggregate income (GNP) and that this imbalance is reflected in a current account deficit.

We can also write (4.2) as:

$$GNP = C + I + G + NX + NFIA$$

$$\text{or, } GNP = C + I + G + CA, \text{ (where } CA = NX + NFIA \text{)}$$

$$\text{or, } GNP - (C + I + G) = CA \quad (4.3)$$

Note that CA represents the ‘current account balance’ of a country, which may be positive (current account surplus) or negative (current account deficit).

The identity (4.3) indicates that aggregate domestic absorption ($C+I+G$) exceeds aggregate income (GNP) in a country that has a current account deficit ($CA < 0$). Such a country is a net debtor (or, borrower) as it has to borrow to bridge the gap between income and expenditure whereas aggregate income exceeds expenditure in a country running a current account surplus ($CA > 0$) and it is a net lender.

A current account deficit may be financed by borrowing from the rest of the world or by sale of foreign assets held by domestic agents. We will have more to say on this in the next section on balance of payments.

The national income identity can be further used to understand whether the real sector imbalance reflected in a current account deficit, stems mainly from the private sector, or the government sector, or both.

From the income side, we know that aggregate income is either consumed (C), or saved (S) or paid out in taxes (T), so that,

$$C + S + T = \text{GNP} \quad (4.4)$$

Combining (4.3) and (4.4), we can write,

$$C + S + T = \text{GNP} = C+I+G+CA$$

$$\text{or, } C + S + T = C+I+G+CA,$$

$$\text{or, } S + T = I+G+CA$$

$$\text{or, } (S - I) + (T - G) = CA \quad (4.5)$$

From identity (4.5) it is clear that a current account deficit ($CA < 0$) reflects either an excess of private investments over private savings (i.e., $(S - I) < 0$) or, a budget deficit (i.e., $(T-G) < 0$) or, both. Also, if $(S - I) > 0$ but, $(T-G) < 0$ and absolute value of $(T-G)$ is larger than that of $(S - I)$, there would be a current account deficit. The identity (4.5) represents the ‘twin deficits’, where an external deficit ($CA < 0$) is reflected as a deficit in the private sector (i.e., $(S - I) < 0$), or in the government sector (i.e., $(T-G) < 0$), or in both.

The national income identities clearly indicate that an imbalance in the current account reflects an imbalance in the real economy. That is, whenever a country runs a current account deficit, there is an excess of aggregate expenditure over income; this may occur because, private savings are too low, or there is a private investment boom, or, the government is running a budget deficit or some combination of these factors. Conversely, aggregate income exceeds expenditure in a country that runs a current account surplus and either its private savings would exceed investments (i.e. $(S - I) > 0$) or, public savings would be positive (i.e., $(T - G) > 0$) or, both.

The main difference between closed and open economies should be evident from the above discussion. In a closed economy, by definition, there are no transactions with the ‘rest of the world’ and CA is zero. Thus aggregate